

Data-Driven Modeling of Urban CO₂ Emissions Using SUMO-Based Traffic Simulation and Machine Learning

Md. Mahmudul Amin Shakil, Monishanker Halder, Rakesh Biswas

Department of Computer Science and Engineering

Jashore University of Science and Technology

Jashore, Bangladesh

ma.shakil330@gmail.com, m.halder@just.edu.bd, 200112.cse@student.just.edu.bd

Abstract—With the rapid growth of population, transportation demand has significantly increased in the urban areas of developing countries like Bangladesh. Major cities are facing transport-related challenges like affordable public transport. Public transport is generally shared by the public. To meet this demand, both cost-efficient and environmentally friendly solutions are needed. In this scenario, battery-powered vehicles like three-wheeled e-bikes, electric rickshaws, and electric vehicles became a part of the urban transportation system as an alternative to the traditional engine vehicles [1]. Recently, the number of battery-powered vehicles has massively increased due to their accessibility and comfort [2]. But these battery-powered vehicles are limited by a short operational range (60–100 km per charge), leading to frequent downtime. The approach involves developing a high-fidelity traffic simulation model of Jashore using SUMO (Simulation of Urban Mobility), incorporating real-world vehicle data. It validates a Vehicle-to-Infrastructure (V2I) data collection architecture that captures dynamic emission data. This dataset will be used to train four machine learning algorithms. Such as linear regression (LR), random forest (RF), gradient boosting (GB), and extreme gradient boosting (XGBoost) to predict how much carbon is emitted in the environment of Jashore City. This software-based strategy is expected to significantly know about carbon emissions from different types of fuel-based vehicles. Among these models, Random Forest performed well with the highest accuracy (93.15%). This research emphasizes the entire life cycle emissions of fuel-based vehicles.

Index Terms—SUMO, Carbon Emissions, Fuel-Based Vehicles, Data-Driven Approach, Machine Learning

I. INTRODUCTION

Global warming has been one of the most burning concerns over the last several decades. The majority of world climate change is mostly dependent on the greenhouse gases (GHGs) like carbon dioxide (CO₂), methane (CH₄), and nitrous oxide (NO) [3]. The effects of climate change that we are facing in every corner of the world are undeniable. Because of climate change, Bangladesh is one of the most vulnerable countries in the world to this global warming [4]. Climate change is an issue that has been of much concern for several years, and greenhouse gas emissions (GHGs) are also the rising concern. The energy, transportation, agriculture, building, waste management, and afforestation and reforestation sectors are the

sources of human-caused emissions. The transportation sector is the main source of globalization for human beings.

The International Energy Agency [5] declares that the transportation sector is the second highest responsible for emitting greenhouse gases (GHGs). 75% of total CO₂ emissions in the environment are produced by the transportation activities [6]. During the pandemic of COVID-19, the emission of CO₂ decreased to 34.4 million tons (MT). This value is lower than the previous year by about 36.1 million tons (MT) [7]. In this research, we use Simulation of Urban Mobility (SUMO). It is an open-source microscopic traffic simulator. It is widely used for road traffic system modeling and estimates the environmental impacts, such as CO₂ emissions. SUMO can simulate the traffic flow and analyze the emissions that are generated by the different types of fuel-based vehicles and road traffic conditions. The effects of greenhouse gas emissions (GHGs) are very harmful for human beings and for the atmosphere. The transportation sector is the most dangerous sector for CO₂ emissions. As a result, it causes many deaths of human lives and animals. Now there is a great concern about global warming; we must find a better solution for reducing the greenhouse gas (GHG) emissions.

II. RELATED WORKS

This study introduces a deep learning approach for forecasting CO emissions from traffic vehicles using deep learning models like Long Short-Term Memory (LSTM) and Bidirectional LSTM (Bi-LSTM). They introduced a vehicular emission dataset, which has many attributes like engine size, fuel type, consumption patterns, etc. For handling the outliers, they used the k-means clustering technique. The Bi-LSTM model performed better than the LSTM model and achieved a high R² score of 93.78% [8], which indicates excellent prediction accuracy. There are some metrics, such as MSE, RMSE, and Pearson's R, that are used to compare the model's performance. This study's main aim is to support planning smart city and environmental policymaking in high-carbon-emission regions.

This research aims to reduce emissions and develop sustainable transport development. For CO₂ emissions, they fo-

cus on the transportation sector. This study applies three deep learning algorithms, like Multi-layer Perceptron (MLP), Extreme Gradient Boosting (XGBoost), and Support Vector Machine (SVM), to predict the CO₂ emissions. The statistical indicators' R_2 value ranges from 0.8969 to 0.9886, and the RMSE value ranges from 0.0333 to 0.1007 [9]. Burning fossil fuel is much more responsible for growing air pollutants in the atmosphere, which worsens the air quality on a large scale. Due to limited resources of energy, the energy demand will be more efficient for future energy resources. This study is restricted to Turkey, which influences the CO₂ emissions in the transportation sector.

This paper focuses on the environmental effect by analyzing the life cycle of CO₂ emissions with machine learning algorithms. The author makes models for estimating emissions through various stages like manufacturing, usage, and disposal. Though EVs are clean, the upstream emissions must be considered. In this study, they used machine learning algorithms like linear regression, random forest, support vector regression, and Bi-LSTM [11] for prediction and accuracy. These models show a better result and effectively predict the life cycle of carbon emissions. But the Bi-LSTM model performs well among the other models. These findings give importance to indirect emissions in sustainable planning.

This paper proposes a deep learning method for predicting vehicular carbon emissions. For accurate prediction, this study maintains environmental data, types of vehicles, and travel patterns. The On-Board Diagnostics-II (OBD-II) dataset is used to assess emission trends using a deep neural network model. They proposed a 2-layer Long Short-Term Memory (LSTM) model [12]. It gives better results and accuracy than traditional machine learning models like linear regression, random forest, gradient boosting, support vector machines, and so on. This model can help in the urban areas with reducing transport-related emissions. This study supports the policy of making AI-driven environmental monitoring.

In this study, the author rates vehicles based on their CO₂ emissions by using supervised machine learning algorithms. Fuel consumption, engine size, and other specifications—they classify the emission levels using these data. Supervised machine learning algorithms are tested for the result and compared. There are about 203 g/km of CO₂ emitted from the new passenger cars. The average CO₂ emissions were about 170 g/km. This research gives a solving direction of aware vehicle practices on roads that should be guided by the consumers and regulators. These models give high accuracy and are suitable for real-time applications. In urban traffic control systems, it also suggests integrating these models [13].

In this paper, the author applies many regression-based machine learning models (linear regression, lasso regression, and ridge regression) for predicting the CO₂ emissions from the vehicles. For training, they used the “CO₂ Emissions Canada.csv” dataset that contains many vehicle specifications. For each algorithm they calculate the accuracy, R-squared, mean absolute percentage error (MAPE), and root mean square error (RMSE) for the comparison. They find an optimal

balance between accuracy and overfitting by using ridge regression [14]. For data-driven policy recommendations in the automobile sector, this research is more helpful.

This paper represents a machine learning-based approach to predicting vehicular carbon emissions using many features like vehicle type, engine specifications, fuel type, weight, driving patterns, environmental conditions, etc. They tested many machine learning algorithms, such as decision trees, random forests, linear regression, neural networks, and support vector machines. In these Random Forest algorithms, the most accurate prediction. This study's focus is to improve environmental impact monitoring through these methods [15]. By integrating these models into automobile industries and environmental agencies, it will be greatly beneficial for all of us. These models also support the vehicular emission certification processes.

This paper discusses futuristic cities like NEOM (Saudi Arabia's smart city project) to predict the carbon emissions using machine learning algorithms. They compare many machine learning algorithms like the LogitBoost ensemble model and decision trees and explore urban parameters for forecasting CO₂ emissions. They analyzed the “Fuel Consumption” dataset from Kaggle. The best accuracy comes from the decision tree technique at 99.7% [16]. About 99.3% accuracy was also achieved by the LogitBoost Ensemble technique. The goal is to help by planning a sustainable infrastructure in the urban areas. This study promotes energy efficiency and forecasting carbon dioxide in urban areas.

In this article, the author's major concern is gradually released greenhouse gas (GHG) emissions. The main victim of global warming is CO₂. They explore a deep learning hybrid model of a Convolutional Neural Network and Long Short-Term Memory Network (CNN-LSTM) [17] to predict the CO₂ emissions, which focuses on India's transportation data. They integrate historical emission trends and weather data for enhanced prediction of carbon emissions. This model gives better accuracy over traditional machine learning models and single deep learning models. This study shows India's growing need for emission control in the urban and rural areas. This study suggests policy-level emissions forecasting by using hybrid AI systems and is also applicable for smart city planning.

III. SIMULATION ENVIRONMENT

In our study, the Simulation of Urban Mobility (SUMO) framework is used. Its graphical network editor (NetEdit) is used to design and simulate vehicles and analyze vehicular emissions of Jashore City. We need to install the SUMO suite for preparing the simulation environment. It includes NetEdit for network creation and Sumo-GUI for graphical simulation. Python is also installed to enable automated route generation and analysis. All the required executable locations were added to the system's path. For this, the overall methodology consists of five primary steps. Such as environment setup, network design, route generation, simulation execution, and data analysis.

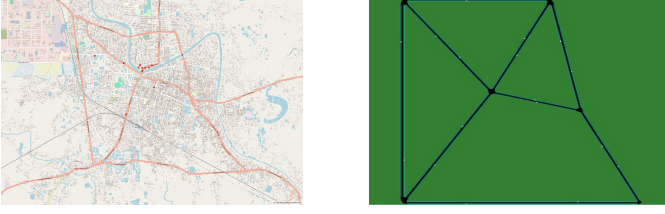


Fig. 1. OSM Map VS Created Map in NetEdit

A. Environment Setup

The simulation environment is created by using the operating system Windows 11 (64-bit); SUMO version 1.19.0 (including SUMO-GUI and NetEdit); and Python version 3.12. Additional tools like netconvert for creating networks are included. All these tools are added to my system path to enable command-line access. The environment setup is much needed for ensuring compatibility among SUMO, Python scripts, and XML file processing.

B. Network Creation in NetEdit

For my study area (Jashore), I created a road network design using NetEdit. For this there are some steps included such as,

1) *Node Creation*: For node creation, I created a total of six nodes for my network defined in the file.nod.xml file. Each node specifies the ID and its location's (x, y) coordinates. For example, `<node id="0" x="400.0" y="1400.0" type="traffic light"/>` indicates the node ID is 0, with location coordinates (400.0, 1400.0) functioning as a traffic light, while `<node id="3" x="1200.0" y="300.0" type="priority"/>` also indicates the node ID is 3, with location coordinates (1200.0, 300.0) functioning as a priority junction. In the total of six nodes, five nodes (Palbari, New Market, Monihar, Chakra, and Doratana) are functioning as a traffic light, and only one node (Muroli Mor) is functioning as a priority junction. In this file all node points are defined in the network [18].

2) *Edge Creation*: There are a total of eighteen edges that are created in the opposite direction in the network. This is defined as the file.edg.xml file. Each edge declares edge ID, the path from one node to another node (indicated by from and to attributes), priority, number of lanes, vehicle speed, and distance between two nodes. For example, `<edge id="a01" from="0" to="1" priority="2" numLanes="1" speed="11.11" distance="2550.00"/>` represents an edge whose ID is a01, connecting node 0 with node 1. This edge declares the priority as 2; two lanes are directed in the same direction as a single edge, the speed limit of vehicles is 11.11 m/s, and the length of the edge is 2550.00 meters [?].

I created the Jashore City map using NetEdit with nodes and edges of these roads. The map has been taken from the OSM (OpenStreetMap). The OSM map and my created map of Jashore City are shown in Fig. 1.

C. Network Generation

For network file creation, we used this command, “netconvert -n file.nod.xml -e file.edg.xml -o file.net.xml,” which automatically generates the network file, and this saves as a file.net.xml file.

D. Route creation

For route creation, I manually create the vehicles' routes using the edges to represent real-world vehicle distributions. In the file.rou.xml file, the routes are specified along with vehicle types (`<vType>`) that include many parameters like vehicle type, vehicle class, acceleration, deceleration, mass, sigma, length, width, height, minimum gap, maximum speed, emission class, lcStrategic, lcCooperative, lcSpeedGain, lcKeepRight, and laneChangeMode. For example, `<routes> vType id="passenger" vClass="passenger" accel="2.6" decel="4.5" mass="1500.0" sigma="0.5" length="4.5" width="2.0" height="1.5" minGap="2.0" maxSpeed="22.0" emissionClass="HBEFA4" lcStrategic="1.0" lcCooperative="1.0" lcSpeedGain="1.0" lcKeepRight="0.3" laneChangeMode="1621"/>` defines the passenger (private car) vehicle type, while `<vehicle id="0" type="passenger" depart="0.00">` and `<route edges="b03 a08 b06 a01"/>` define the vehicle with id=0, vehicle type=passenger, vehicle departing at time 0.00, and following the given sequence of edges [19]. A total of ten types of different fuel-based vehicles are defined by their parameter values as follows. These parameters for the different types of vehicles are defined in the file.rou.xml named file.

IV. DATASET DESCRIPTION

Data extraction was done by parsing the vehicle-emissions.csv file using Python's TraCI integration. There is a total of ten types of fuel-based vehicles, like passenger cars, buses, trucks, taxis, emergency vehicles, bikes, delivery vehicles, coaches, SUVs, and trailers. About hundreds of routes are created with different route edges for these different types of vehicles. All routes covered the most congested areas of Jashore City. The total emissions from these different types of vehicles are calculated by these distances. By running the vehicles, they emit many greenhouse gases (GHGs) like CO₂, CO, HC, NO_x, and PM_x. In data processing, this summarizes the per-time-step statistics for each vehicle type's emissions. Results are grouped by the vehicle class to compare the emissions across all the modes. We visualize all types of vehicles running in the streets, and some vehicles are waiting in the traffic. Variation of the traffic demand: this comparison was made, and it assesses the environmental impacts on the vehicle-type distributions. This dataset is used to train different machine learning models. Linear regression, random forest regressor, gradient boosting, and XGBoost models performed a comparative study to find out the best model for accurate predictions.

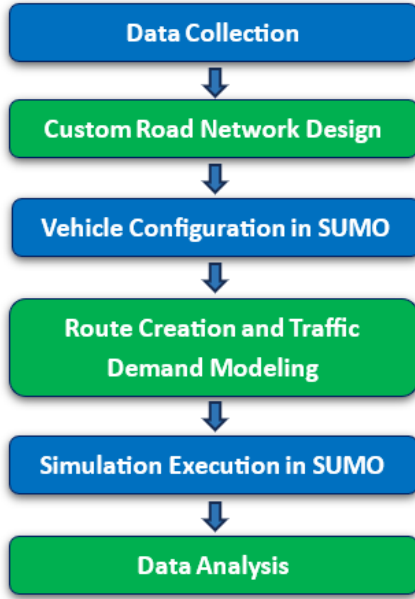


Fig. 2. Proposed Research Framework

V. METHODOLOGY

This research follows a sequential, modular design that integrates simulation, real-time data logging, and data analysis. The flowchart 3.1 below illustrates the step-by-step approach used to develop and evaluate carbon emissions from fuel-based vehicles in Jashore City. The research design flow of this work is shown in Fig. 2.

After simulation, the dataset has been created. The next step is to develop some different machine learning models like linear regression, random forest regressor, gradient boosting, and XGBoost. We evaluate these models to predict the emissions produced by the different types of fuel-based vehicles. In this paper, we consider these four models to compare accuracies, F1 scores, R^2 , MAEs, MSE, RMSE values, and the capability to fit with the created dataset.

A. Linear Regression (LR)

Linear regression model is a linear relationship between the independent variables and the target variables. In my dataset, vehicle features are independent variables, and CO_2 -mg is my target variable. It assumes that:

$$CO_2 = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p + \varepsilon \quad (1)$$

Where p = number of features, x_j = independent variables, β_j = regression coefficients

This equation finds coefficients that minimize the mean squared error (MSE) values. The CO_2 emission changes linearly with the vehicle features. It is fast and simple with linear relationships. It works well with this dataset. The accuracy of this model is about 92.72%. It gives better results. In this

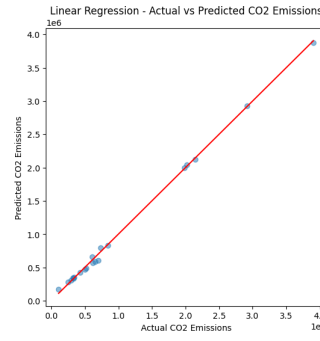


Fig. 3. Linear Regression Model

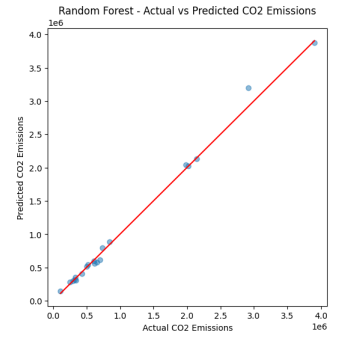


Fig. 4. Random Forest Model

model the random state is 42 and the test size is 0.2, which is pipelined to maximize the output data. It is a good baseline model for prediction. About a 0.9984 R^2 value in this model. The actual vs. predicted CO_2 emissions graph for the linear regression model is shown in Fig. 3.

B. Random Forest (RF)

A random forest regressor is commonly used in supervised machine learning models that take the output from multiple decision trees. It averages the prediction values to get the final output. In this model, the random state is 42, max depth is 10, and total estimators are 200. This model gives better accuracy than the linear regression model. About 93.15% of accuracy comes from this model. In my dataset, this model captures the nonlinear relationships between vehicle parameters and CO_2 -mg parameters. This model performs well with many features and handles complex interactions automatically. The R^2 value is about 0.9944 in this model. Random forest regressors usually perform well with emission datasets. The actual vs. predicted CO_2 emissions graph for the random forest model is shown in Fig. 4.

C. Gradient Boosting (GB)

Gradient boosting algorithms sequentially build trees. The previous trees' errors are corrected by each new tree. It learns patterns gradually and optimizes loss function using gradient descent. This model is more sensitive to hyperparameters than the random forest model. In this model, the total number of estimators is 200, the learning rate is 0.05, max depth is 5, and the random state is 42. It gives much better accuracy. About 92.66% accuracy comes from this model. It produces much better accuracy and performance. The actual vs. predicted CO_2 emissions graph for the gradient boosting model is shown in Fig. 5.

D. Extreme Gradient Boosting (XGBoost)

The XGBoost (Extreme Gradient Boosting) algorithm is an advanced version of the gradient boosting algorithm. For better performance we established this algorithm. It uses regularization, parallel processing, and true pruning and handles missing values automatically in this dataset. Almost 92.90% of accuracy comes from this model. This accuracy is the second

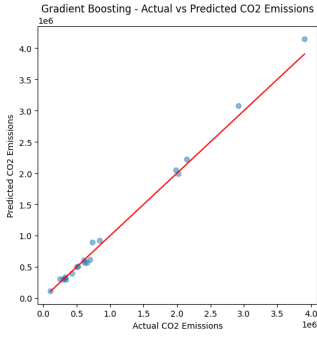


Fig. 5. Gradient Boosting Model

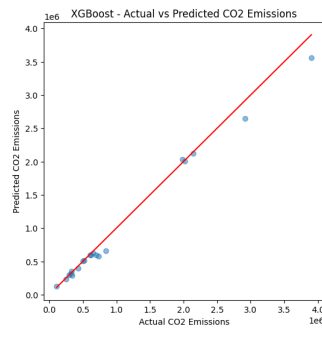


Fig. 6. XGBoost Model

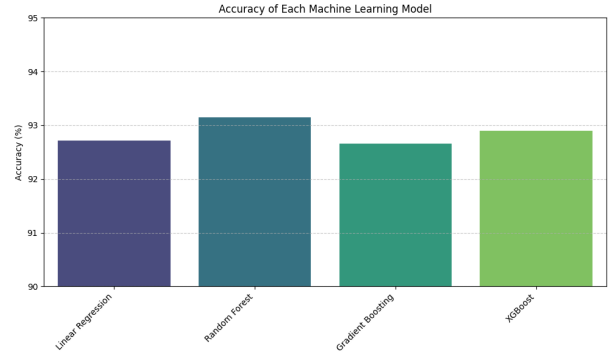


Fig. 7. Accuracy Comparison Among Each ML Models

TABLE I
COMPARISON OF MACHINE LEARNING MODELS

Model	R ²	MAE	MSE	RMSE	F1 Score
Linear Regression	0.9984	31643.81	1.62E9	40258.94	1.0
Random Forest	0.9944	45829.49	5.63E9	75023.27	1.0
Gradient Boosting	0.9923	61220.40	7.71E9	87843.07	1.0
XGBoost	0.9862	68865.49	1.38E10	117397.8	0.95

highest accuracy value among these four models. RMSE values are higher in this model than in the other models. XGBoost is very efficient for large datasets and handles complex vehicular emission patterns. The actual vs. predicted CO₂ emissions graph for the XGBoost model is shown in Fig. 6.

VI. RESULTS

This chapter presents the quantitative results obtained from the simulation of the vehicle-to-infrastructure (V2I) data collection system. It provides a detailed analysis of these results and discusses their significance in the context of the research objectives. The total travel time for all types of vehicles to finish their route is about 2853 seconds. The most critical finding is the successful measurement of carbon emissions, which is a direct result of configuring vehicles accurately according to the parameters in SUMO. The collected dataset enables the calculation of crucial statistics of vehicle CO₂, which are essential inputs for the optimization of machine learning algorithms.

We established four machine learning algorithms—linear regression, random forest regressor, gradient boosting, and XGBoost. Among these models, we compared R², MAE, MSE, RMSE, and F1 Score. In the linear regression model, the R² value is higher than other models. Both MAE and RMSE values are high for the XGBoost algorithm. These values are compared in Table I for the most accurate predictions of these models. The most accuracy comes from the random forest regressor model. About 93.15% accuracy in this model, which is higher than the other models. With the limitations and barriers, the random forest regressor model is the best fit for the accurate prediction. The comparison of accuracy for each model is shown in Fig. 7.

The heatmap of RMSE (Root Mean Square Error) comparison visually represents the prediction error magnitude. This comparison is among the machine learning models for CO₂ emissions estimation. The RMSE value is measured by the average deviation between the actual CO₂ emissions values and the model-predicted CO₂ emissions values:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (2)$$

Where y_i = actual value, $\hat{y}_i$ = predicted value, n = total number of samples

A lower RMSE value gives better prediction accuracy. So, the linear regression model gives the better prediction accuracy as the value of RMSE is lower than the other models.

TABLE II
DATA EXTRACTION OF EMISSION FILE (PARTIAL EXAMPLE)

vehicle_id	vehicle_type	depart_time_s	travel_time_s	CO ₂ _mg
0	passenger	1	214	432883.72
1	bus	2	101	186549.67
2	truck	3	169	319260.18
3	bus	4	104	217430.99
4	taxi	5	132	250753.27
5	trailer	6	258	505656.34
6	emergency	7	229	449420.69
7	taxi	8	170	336957.45
8	bus	9	2853	4404468.05
9	trailer	10	1374	2122456.68

In Table II, these emission values comes through the various types of fuel-based vehicles. This is the partial example of my dataset.

The correlation heatmap is another term that is very important. It shows the strength of linear relationships between the actual CO₂ emissions values and the model-predicted CO₂ emissions values. The correlation heatmap between actual and predicted CO₂ emissions is shown in Fig. 8. This correlation is based on the Pearson correlation coefficient:

$$r = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \quad (3)$$

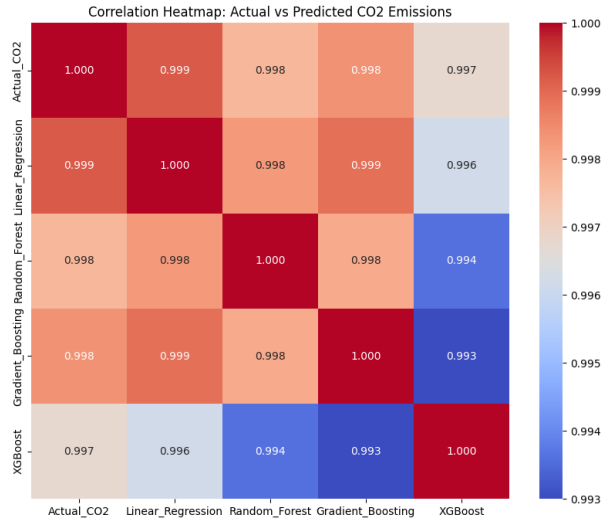


Fig. 8. Correlation Heatmap Between Actual and Predicted CO₂ Emissions

Where $\text{Cov}(X, Y)$ = covariance, $\sigma_X \sigma_Y$ = standard deviations

VII. CONCLUSIONS

This research successfully designed, implemented, and validated a robust vehicle-to-infrastructure (V2I) data collection architecture utilizing the Eclipse SUMO simulation environment. The core objectives were achieved through the creation of a custom road network of Jashore, Bangladesh; a realistic configuration of ten fuel-based vehicles (like passenger, bus, truck, taxi, emergency, bike, delivery, coach, SUV, and trailer); and the establishment of a functional data pipeline. The key successes of this study include V2I architecture validation, complete coverage and redundancy, and data-driven insights. This chapter summarizes the key achievements of the research and outlines the limitations encountered during the implementation and simulation phase. Specifically focusing on the application of machine learning algorithms for optimizing carbon emissions. Four machine learning algorithms were employed to compare their performance and find out the best model among them for our created dataset. Among these models, the random forest performs well with the highest accuracy. It is the most suitable choice of model for our analysis.

In this paper, the road network is limited to the custom creation of only six major junctions of Jashore City, connected by eighteen road edges. It configures all vehicles to depart simultaneously at 0.00s. This setup ensures comparative analysis but does not fully represent the dynamic, stochastic nature of real-world traffic flow. Future research could be done by including broad real-world data of different regions. It will expand the dataset and improve the machine learning models' performance. It is also compared with the regional differences of carbon emissions from different fuel-based vehicles. For more accurate predictions, there need to be established advanced machine learning models like LSTM. This study

contributes to a good understanding of the environmental carbon emissions from fuel-based vehicles. It also promotes more sustainable decision-making in the transportation sector in our country.

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